PRATEEK BALIYAN

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LinkedIn | GitHub | LeetCode | Portfolio

SUMMARY

B.Tech CSE (AI) student focused on **AI/ML, Computer Vision, Quantitative Analytics and Full-Stack Development**, experienced in building end-to-end applications using **Python, C++, FastAPI, Next.js, React, Node.js, MongoDB, PostgreSQL and Docker**.

EDUCATION

KIET Group of Institutions, Ghaziabad **2024–2028**
B.Tech in Computer Science and Engineering (Artificial Intelligence) **CGPA: 8.38/10**
The Skyland School – Class XII, CBSE **94.6% | 2024**
The Skyland School – Class X, CBSE **95.4% | 2022**

PROJECTS

VeriStream – Real-Time Deepfake Detection & Forensic Interceptor GitHub | Live

- Built a real-time facial authenticity system using **Python, FastAPI, WebSockets, OpenCV, NumPy, SciPy** and Next.js.
- Combined **rPPG, corneal illumination disparity, facial landmarks/AU micro-tremors, C2PA provenance** and SynthID-related inspection using adaptive sensor fusion and temporal smoothing.
- Generated tamper-evident forensic PDF dossiers using **HMAC-SHA256** with an ISO/IEC 30107-3 PAD-oriented workflow.

DocuAudit – Digital Document Forensics Platform GitHub | Live

- Developed a document tampering detection platform using **Python, FastAPI, OpenCV, PyMuPDF, Scikit-learn, Next.js** and TypeScript.
- Implemented **ELA, noise residuals, ORB + DBSCAN copy-move detection**, PDF revision analysis, font anomalies and PKCS#7 ByteRange validation.
- Built SHA-256 chain-of-custody certificates, checksum validators and an interactive forensic viewer with zoom, magnification and heatmaps.

LiquidityPulse – Real-Time Quantitative Market Analytics GitHub | Live

- Built a real-time market microstructure platform using **Python, FastAPI, WebSockets, Next.js, NumPy, Pandas, Redis** and PostgreSQL.
- Implemented **L2 order books, OFI, VPIN, micro-price** and anomaly detection for spoofing, layering and iceberg activity.
- Added tick-level replay, backtesting, slippage/latency modeling and transaction-cost analysis with interactive market visualization.

AI Productivity Tracker GitHub | Live

- Developed a full-stack productivity analytics system using **React, Node.js, Express, MongoDB, JWT** and a Chrome extension for browser activity tracking.
- Added productive/distraction classification, productivity scoring, streaks, daily/weekly analytics and leaderboard-based behavior insights.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL **AI/ML:** Machine Learning, Computer Vision, Scikit-learn, NumPy, Pandas
Frontend: React.js, Next.js, HTML, CSS, Tailwind CSS **Backend:** Node.js, Express.js, FastAPI, REST APIs, JWT
Databases: MongoDB, PostgreSQL **Core CS:** DSA, OOP, DBMS, Operating Systems, Computer Networks
Tools: Git, GitHub, Docker, Postman, VS Code, Vercel, Render

ACHIEVEMENTS & CERTIFICATIONS

AWS Certified Cloud Practitioner | **182+ LeetCode Problems** | Built and deployed multiple AI/ML and full-stack applications.

ADDITIONAL PROJECTS

Location Intelligence: React, Vite, Express.js, OpenStreetMap APIs; interactive maps, heatmaps, business discovery and location opportunity analysis.
ML Projects: Heart Disease Prediction, Bulldozer Price Prediction, Internet Usage Clustering, Naive Bayes Weather Prediction, COVID-19 Forecasting.